1. Data mining is the process of discovering interesting patterns, anomalies trends and useful knowledge from large volumes of data stored in databases data warhouses or other repositories.

Tasks:

1. classification
2. clustering

2. Data cleaning is the process of detecting and correcting corrupt, inaccurate, incomplete or irrelevend records from a dataset to prepare it for analysis.

data quality issues:

1. Missing values: Attributes with missing or empty values for certain records.
2. Noisy data: Data containing errors, outliers or random variance that distorts real patterns.

3. Data integeration is the process of combining data from multiple heterogenous sources into a unified and coherent data store.

challenges

① Entity identification problem

② Redundancy and data value conflicts

4. what is covariance analysis? list the 2 covariance formulas.

covariance analysis measures how two numerical variabler change together. A positive covariance indicates that both variables tend to increase together, while a negative covariance indicates an inverse relationship.

$$Cov(x,y) = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

Q5. Suppose stock A & B

mean of stock A ($\bar{x}$)

$$\frac{2+3+5+4+6}{5} = \frac{20}{5} = 4$$

mean of stock B ($\bar{y}$)

$$\frac{5+8+10+11+14}{5} = \frac{48}{5} = 9.6$$

$(2-4)(5-9.6) = 9.2$
$(3-4)(8-9.6) = 1.6$
$(5-4)(10-9.6) = 0.4$
$(6-4)(14-9.6) = 0$
$(4-4)(11-9.6) = 8.8$

$9.2+1.6+0.4+0+8.8 = 20$

$$\frac{20}{5-1} = 5$$

$Cov(A, B) = 5 > 0$

positively correlated & are not independent.

6. GUI

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Pattern evaluation

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Data Mining Engine

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Database / Data warehouse server

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Data cleaning, Integration & selection

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Data warehouse

7. $\bar{x} = \dfrac{4+5+6+7+9}{5} = \dfrac{31}{5} = 6.2$

$\bar{y} = \dfrac{8+3+7+5+6}{5} = \dfrac{29}{5} = 5.8$

$(4-6.2)(8-5.8) = -4.84$

$(5-6.2)(3-5.8) = 3.36$

$(6-6.2)(7-5.8) = 0.24$

$(7-6.2)(5-5.8) = 0.64$

$(9-6.2)(6-5.8) = 0.56$

$= \dfrac{-1.8}{5-1} = -0.45$

8. Data cleaning:

fills missing values, smooths noisy data, and remove outliers and inconsistent date.

Data integration: Merges data from heterogenous sources into a unified structure.

Data reduction: obtains a reduced representation of the equivalent analytical results.

Data transformation: converts data into formats suitable for mining via normalization aggregation or binning.

9. Bin 1 (9 values)

13, 15, 16, 16, 19, 20, 20, 21, 22

Bin 2

22, 25, 25, 25, 30, 33, 33, 35

Bin 3

35, 35, 35, 36, 40, 45, 46, 52, 70

10. Data discretization transforms continous numerical attributes into categorial interval concepts or nominal labels.
Redues data complexity, handles continious features in algorithms requiring discrete input and speeds up computational mining Process